



Answer any five questions of the followings.

5 × 6 = 30

- 1(a). Define the basic data types of C programming with typical memory requirements. CLO1 2
- 1(b). Explain all formats of integer data types with memory size and format specifiers. CLO1 4
- 2(a). Distinguish between Structured Programming Languages and Object Oriented Programming. CLO3 2
- 2(b). Briefly explain compilation process of Structured Programming Language. CLO3 4
- 3(a). Differentiate between compiler and interpreter. CLO5 2
- 3(b). Fill in the following blanks: CLO3 1x4=4
- The _____ standard library function displays information on the screen.
 - The _____ selection statement is used to execute one action when a condition is true and another action when that condition is false.
 - In counter-controlled iteration, a(n) _____ is used to count the number of times a group of instructions should be repeated.
 - The number used to refer to a particular array element is called its _____.

4(a). Consider the following program:

CLO4 2

```
#include <stdio.h>
void main()
{
    int k;
    for (k = -3; k < -5; k++)
        printf("Hello");
}
```

Re-write the given program using do...while loop and write the output.

- 4(b). Write a complete C program that takes two suitable integer matrices (not necessarily square matrices) as inputs, add them and display the resultant matrix as the output. The sizes and elements of the matrices are to be taken as inputs. Use necessary prompt while taking inputs. CLO4 4

and necessary indentations while showing outputs.

5(a). What is type casting in C?

CLO1 2

5(b). Identify and correct the errors in each of the following statements:

CLO5 1x4=4

- i. `printf("The value is %d\n", &number);`
- ii. `scanf("%d%d", &number1, number2);`
- iii. `if (c < 7); {`
 `prints("C is less than 7);`
 `}`
- iv. `if (c ==> 7) {`
 `print("C is greater than or equal to 7")`
 `}`

6(a). Write the nested loop for the following pattern:

CLO4 2

				5
			4	4
		3	3	3
	2	2	2	2
1	1	1	1	1

6(b). A Palindrome number is a number that remains the same when its digits are reversed, such as 121, 151 etc. Write a program in C to find all the Palindrome numbers from 1 to 500.

CLO4 4

7(a). Write a complete program in C for the following states:

CLO4 6

- i. Read `n` (size of elements) numbers of an array
- ii. Find the minimum and maximum value of the inputted array
- iii. Find the addition of minimum and maximum and check if it is prime or not.



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology

Department of Computer Science and Engineering

EEE 127: Electronic Devices and Circuits

Batch: 25th

Date: 30/06/2024

Mid-Term Examination

Semester: Spring 2024

Time: 1 Hour 30 minutes

Total Marks: 30

28-8-24

Answer any five questions of the following.

5 × 6 = 30

1. (a) Write the relationship equations between α and β in transistor? Explain with proper figure. CLO2 2
- (b) Explain band theory with the necessary diagrams. CLO1 4

2.

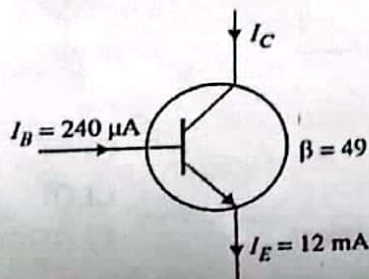


Figure 1

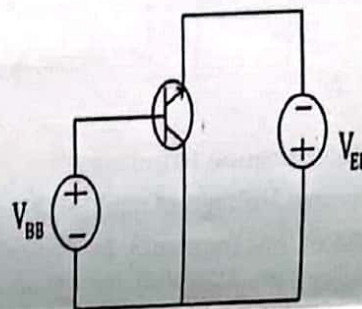


Figure 2

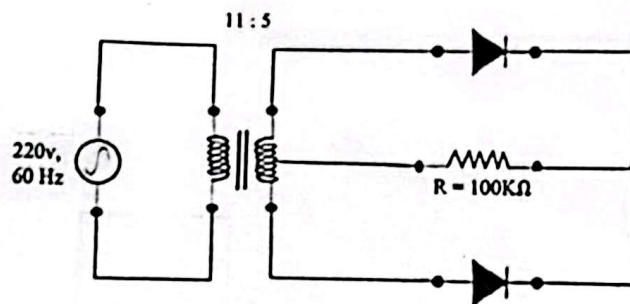
- (a) Find the α rating of the transistor shown in the figure-1. Hence determine the value of I_C using both α and β rating of the transistor. CLO1 3
- (b) From figure-2 calculate I_E in a transistor for which $\beta = 50$ and $I_B = 20 \mu A$. CLO1 3
3. (a) Find the value of β if (i) $\alpha = 0.9$ (ii) $\alpha = 0.98$ (iii) $\alpha = 0.99$. CLO1 2
- (b) Describe three types of transistor configuration with appropriate figures. CLO1 4

4. (a) Explain the I-V Curve of p-n junction for both Forward and Reverse Bias. CLO1 3
- (b) Derive the Ripple Factor for Full Wave Bridge Rectifier. CLO3 3

$$\frac{V_m}{\sqrt{2}} + \frac{\pi}{2V_m}$$

5. (a) What is a half wave rectifier? Explain with appropriate figure. CLO1 2
 (b) Make a comparative analysis between Bridge rectifier and Center-tapped rectifier. CLO1 4

6. (a) What is Peak Inverse Voltage (PIV)? Why is it important in rectification? CLO1 2
 (b) Measure the Peak Inverse Voltage of both Half Wave rectifier and Full Wave rectifier and make comparative analysis. CLO2 4



- (a) What is the Rectification Efficiency? CLO1 1
 (b) Find Peak Inverse Voltage of each diode. CLO1 1
 (c) Determine the output frequency by providing a logical explanation and a diagram. CLO1 1
 (d) Find the output voltage and ampere across the register R. CLO1 3

Hamdard University Bangladesh

Faculty of Science, Engineering and Technology

Midterm Examination (Spring'24)

PHY 111 Electricity & Magnetism/ PHY 121 Physics II/

PHY 121 Physics II (Mechanics, Electricity, Magnetism and Modern Physics)

Time: 1.5 hours

Marks: 30

Answer any 3 (Three) of the following sets of questions

- 1/ (a) State Coulomb's Law. Prove it using Gauss' Law. 5 CLO1
- (b) Four equal point charges, 3×10^{-6} C, are placed at the four corners of a square that is 40 cm on a side. Find the force on any one of the charges. 5 CLO4
- 2/ (a) Define Electric Potential. Explain the equipotential surface with the necessary figures and equation. 4 CLO1
- (b) Determine the electric field due to uniformly charged wire by using Gauss' Law or Coulomb's law. State the reason behind selecting the process. 6 CLO1
- 3/ (a) Derive the equation for Electric Potential for a point charge. 5 CLO4
- (b) What is the electric potential energy of the charge configuration in the figure below? Given, $q_1 = + 1 \times 10^{-8}$ C, $q_2 = - 2 \times 10^{-8}$ C, $q_3 = + 3 \times 10^{-8}$ C, $q_4 = + 2 \times 10^{-8}$ C, and $a = 1$ m. 5 CLO1

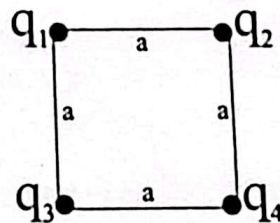


Fig.3. (b)

- 4 (a) What is an electrical dipole? Prove that $U = -PE (\cos \theta - \cos \theta_0)$ (where each symbol used for usual meaning) 6 CLO1
- (b) An electric dipole consists of two opposite charges of magnitude 2 micro coulomb, separated by a distance $l = 1$ cm. It is placed in an external electric field of 2×10^5 N/C. What maximum torque does the field exert on the dipole? 4 CLC



HAMDARD UNIVERSITY BANGLADESH

Department of Computer Science & Engineering
Faculty of Science, Engineering and Technology
BSC in Computer Science & Engineering
ENG 127: English Skills
Midterm Examination
Spring 2024

25th Batch
08 June 2024

2nd Semester
Time: 1 hour 30 minutes

Total Marks: 30

CLOs	Level	Marks
CLO1,2	Understand	6x1=6

Section A

Read the following passage and answer questions 1 - 6:

Cotton is a subtropical plant that grows in many warm areas of the world. It started out as a plant of the tropics but today it is grown in other warm areas that have at least 200 frost-free days. The most important cotton-growing countries are the USA, China, India, Pakistan and Australia. Cotton is a plant that produces fibers, which are used to make clothes and other products, like towels, carpets or sheets. Clothes made out of cotton are especially light and comfortable. Every part of the cotton plant can be used. The long cotton fibers are used to make cloth, the short fibers can be used in the paper industry. You can make oil or margarine out of the seeds of the cotton plant. The leaves and stalks of the cotton plant are plowed into the ground to make the soil better. Other parts of the plant are fed to animals.

1. Which of the following can be inferred from the passage?

- a) High quality towels are produced in Australia. X
- ☒ b) Cotton favors humid and cold days.
- c) Tropical climates are best to grow cotton.
- d) Paper could be purchased very cheaply in bulk in India X

2. Only a few parts of the cotton plant can be used in industry.

- ☒ a) True b) False c) Not stated

3. "made out of" (Clothes made out of cotton are especially light and comfortable) is similar in meaning to -

- a) Shaped by ☒ b) resulted in ☒ c) manufactured to d) produced from
4. It is understood that in Scandinavian countries, it is impossible to grow cotton.

- a) True b) False c) Not stated

5. We can clearly understand that -

- a) Cotton can be grown in any cold place with more than 200 frost free days.
- b) Long cotton fibers are used to feed animals X
- ☒ c) China is one of the cotton exporting countries
- d) Cotton has a variety of uses in industrial production

6. You can make oil or margarine out of the fiber of the cotton plant.

- a) True ☒ b) False c) Not stated

7. Fill the gaps with suitable words from the box:

exceptionally	margarine	Leading	cold	manufacture	edible	stalks
---------------	-----------	---------	------	-------------	--------	--------

The USA, China, India, Pakistan, and Australia are the a).....cotton-growing nations. The fibers from the cotton plant are used to make clothing as well as other items like sheets, towels, and carpets. Cotton clothing is b)..... comfy and light. You can use every part of the cotton plant. While the short cotton fibers can be used in the paper industry, the long cotton fibers are used to c)..... textiles. Cotton plant seeds can be used to manufacture d)..... or oil. To improve the soil, cotton plant leaves and e)..... are mashed into the ground. Animals are fed plant components that are not f).....

CLO1 Apply 6x1=6

Read the following passage and answer questions 8- 12:

Trade happens because people need or want things that they don't have. We also trade for work that we cannot do ourselves. Trade between countries happens for the same reason. Some countries, for example, have natural resources, like coal, oil or wood which other countries might want to buy. They try to sell the goods, products or services that they have too much than the other countries. They earn money from these sales and then can buy the things that they themselves need and cannot produce on their own. Both producers and consumers profit from international trade. If countries can produce goods more cheaply than others because they specialize on them why not sell them to other countries. They make more money on one side and consumers in other countries can buy goods that are cheaper. Even though many nations have a lot of different goods to export there are countries that depend only on one or two products to get money. Saudi Arabia, Kuwait and other countries of the Middle East depend on oil exports, because it is pretty much the only thing that they can sell. Poor countries in Africa depend largely on the export of tropical farm products to get money.

8. Some countries sell products-
- only when they have more than they need in two years
 - to get oil and farming equipment
 - to make money to buy some others they need
 - that they do not need any more in their lands
9. Which of the following is NOT true?
- If we cannot provide services on our own, we do outsource them to others.
 - International trade paves the way for tax avoidance.
 - Kuwait is one of the top oil exporters.
 - Some African countries rely mainly on farm products.
10. Trade has come into existence as people need to exchange goods and services.
- False
 - Not stated
 - True
11. We can infer from the passage that -
- the rich countries are those who only have oil reserves
 - some poor African nations get little money by exporting
 - all poor countries produce and export farm products
 - some countries import many products but not oil and coal
12. It is more preferable to be consumers than producers.
- False
 - True
 - Not stated

13. Answer the following questions from the above passage:

CLO1,2 Understand 5x1=5

- Why do people need to trade?
- Who get profit from international trade?
- What did the author say about the middle east countries?
- What is the author's purpose of writing the above passage?
- Why do some countries sell the products that they specialize in?

Section B

Answer any two of the following:

CLO1,3 Understand 4x2=8

- How to improve speaking skill in English? explain briefly.
- What is extensive reading? What are the differences between extensive reading and intensive reading?
- Briefly discuss some strategies of reading.
- Define scanning and skimming. Which strategy do you prefer most and why?

UNIVERSITY BANGLADESH
Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
B.Sc. in Computer Science and Engineering
MAT 129: Linear Algebra and Complex Variable
Mid-Term Examination
Spring 2024

25th Batch
 24 June 2024

2nd Semester
 Time: 1 Hour 30 Minutes

Answer any 3 (Three) from the following sets of questions

[The figures in the right margin indicate full marks. Symbols used have their usual meaning.]

Sl.No.	Questions	CLOs	Marks
1.	(a) Define homogeneous system of linear equation.	CLO1	01
	(b) Solve the following system of linear equations by row echelon form: $\begin{aligned} x_1 + x_2 + 2x_3 &= 8 \\ -x_1 - 2x_2 + 3x_3 &= 1 \\ 3x_1 - 7x_2 + 4x_3 &= 10 \end{aligned}$	CLO3	03
	(c) Solve the following homogeneous system of linear equations by using Gauss-Jordan elimination method. $\begin{aligned} 2x_1 + 2x_2 - x_3 + x_5 &= 0 \\ -x_1 - x_2 + 2x_3 - 3x_4 + x_5 &= 0 \\ x_1 + x_2 - 2x_3 - x_5 &= 0 \end{aligned}$	CLO3	06
2.	(a) What is inverse of a matrix?	CLO1	01
	(b) Find the solution of linear system by using inverse matrix, the following equations: $\begin{aligned} x_1 + 2x_2 + 3x_3 &= 5 \\ 2x_1 + 5 + 3x_3 &= 3 \\ x_1 + 8x_3 &= 17 \end{aligned}$	CLO3	03
	(c) Find the inverse of the following matrix by using row canonical form: $A = \begin{bmatrix} 3 & 4 & -1 \\ 1 & 0 & 3 \\ 2 & 5 & -4 \end{bmatrix}$	CLO3	06
3.	(a) Define vector vector space.	CLO1	01
	(b) Show that $S = \{(a, 0, c) : a, c \in \mathbb{R}\}$ is a subspace of the vector space $\mathbb{R}^3$	CLO2	03
	(c) If the vectors are $v_1 = (2, 1, 4)$, $v_2 = (1, -1, 3)$ and $v_3 = (3, 2, 5)$ in $\mathbb{R}^3$. Show that $V = (5, 9, 5)$ is a linear combination of v_1, v_2 and v_3 .	CLO3	06
4.	(a) What is periodic matrix?	CLO1	01
	(b) Show that the matrix $A = \begin{bmatrix} 1 & -2 & -6 \\ -3 & 2 & 9 \\ 2 & 0 & -3 \end{bmatrix}$ is periodic.	CLO2	03
	(c) Show that every square matrix can be uniquely expressed as the sum of a symmetric matrix and a skew-symmetric matrix.	CLO3	06
5.	(a) What is rank of a matrix?	CLO1	01
	(b) Find the rank of the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$	CLO3	03
	(c) Find the Eigenvalue and Eigenvectors of the following matrix : $A = \begin{bmatrix} 1 & 2 & -1 \\ 0 & -2 & 0 \\ 0 & -3 & 2 \end{bmatrix}$	CLO3	06

Hamdard University Bangladesh

Faculty of Science, Engineering and Technology

Final Examination (Spring'24)

PHY 111 Electricity & Magnetism/ PHY 121 Physics II/

PHY 121 Physics II (Mechanics, Electricity, Magnetism and Modern Physics)

Time: 2.0 hours

Marks: 40

Answer any 4 (Four) of the following sets of questions

- ✓ (a) Show that magnetic force on a current carrying conductor is $F = i(l \times B)$ (where each symbol is used for usual meaning) 6 CLO2
- (b) What is the force on a singly charged Carbon ion moving with a speed of 3×10^5 m/s at right angles to magnetic field of 7500 G? What is the radius of the circle in which the ion moves? (where charge $q = 1.6 \times 10^{-19}$ C and mass of carbon ion is 19.9×10^{-27} kg) 4 CLO4

- ✓ (a) Define the following term 6 CLO1
- i) Lorentz Force
 - ii) Self Inductance
 - iii) Mutual Inductance
- (b) Find the force on each segment of the wire shown in the figure 2(b), if $B = 0.5$ T. Assume that the current in the wire is 5 A. 4 CLO4

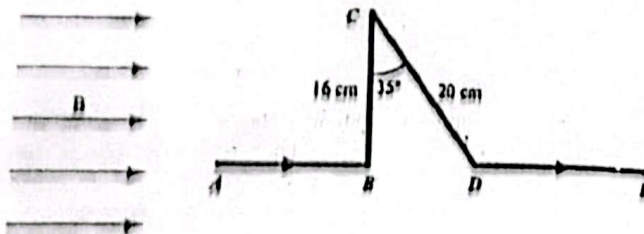
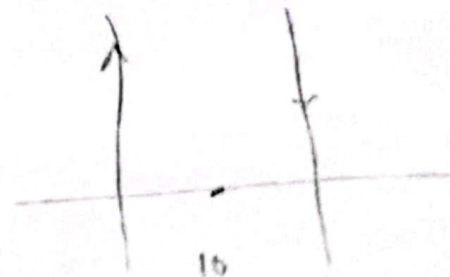


Fig.2.(b)

$$dB \propto \frac{100 \sin \theta}{r^2}$$



- ✓ (a) State and explain Biot-Savart Law with necessary figure. 6 CLO1
- (b) Two long and fixed parallel wires A and B are 10 cm apart in air and carry currents of 40 A and 20 A respectively in opposite direction. Determine the resultant flux density on a line midway between the wires and parallel to them. 4 CLO3

- 4 (a) "Parallel current attract and anti-parallel current repels" – Justify the statement with necessary figure. 6 CLO2
- (b) State Ampere's Law. Determine magnetic field due to a long straight wire applying Ampere's Law (or show that $B = \frac{\mu_0 i}{2\pi r}$). 4 CLO4
- 5 (a) State and Explain Faraday's Law. Also mention the ways by which flux linked with a loop may be changed. 6 CLO1
- (b) The flux linking an air-cored coil of 500 turns changes from $30\mu\text{Wb}$ to $60\mu\text{Wb}$ in 2ms. Calculate the value of the emf induced in the coil. 4 CLO4
- 6 (a) Plot the graph of the growth and decay of current in a RL circuit with respect to time. 6 CLO1
- (b) Determine equivalent energy in Mev of the rest mass of an electron. 4 CLO4



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
EEE 127: Electronic Devices and Circuits

$V_{max} = IR + V_z$

Batch: 25th

Date: 02/10/2024

Final Examination

Semester: Spring 2024

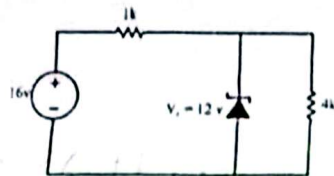
Time: 2 Hours

Total Marks: 40

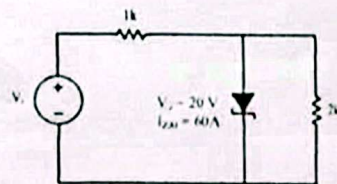
Answer any five questions of the following.

5 × 8 = 40

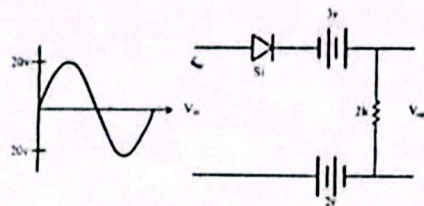
- 1/ (a) Explain the working principle of a Zener diode. CLO1 4
(b) Determine the current across Zener diode and current across the load resistor. CLO2 4



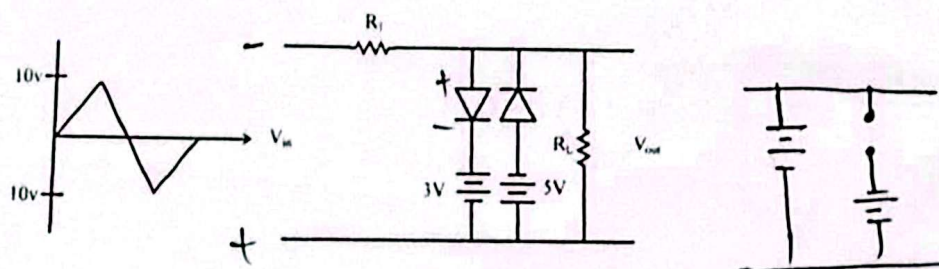
- 2/ (a) How does a Zener diode differ from a regular diode in terms of operation and application? CLO1 4
(b) Determine the range of values of V_i that will maintain the Zener diode in "ON" state. CLO2 4



3. (a) Determine the Output signal of the following circuit: CLO2 4

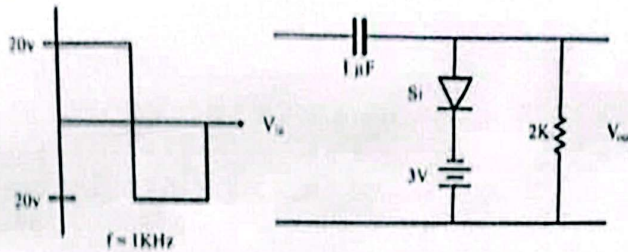


- (b) Draw the Output waveform as well as the Transfer Characteristics of the following circuit: CLO2 4



4. (a) How does the capacitor value affect the performance of the clamper circuit? CLO2 3

(b) CLO2 5

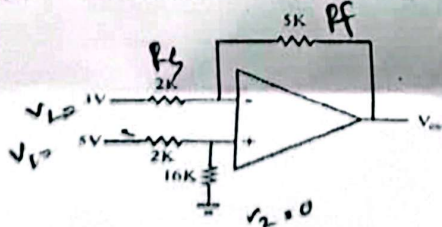


- I. Determine the Output of the above circuit.
- II. Discuss is the capacitance is enough to make the proper output.

5. (a) Explain the concept of "virtual ground" in an ideal op-amp. CLO3 2

(b) Design a circuit with Op-Amp which can average input signal. CLO3 3

(c) Determine the output voltage of the following circuit: CLO3 3



$$V_o = \left(1 + \frac{R_f}{R_s}\right) \times 5$$

6. (a) Make a comparative analysis between Inverting and Non-Inverting Op-Amp. CLO3 2

(b) Design a circuit that outputs the integration of the input signal. CLO3 3

(c) How Op-Amp can be used as Window Comparator? CLO3 3

7. (a) Differentiate between BJTs and FET in terms of construction, operation, and applications. CLO1 2

(b) What is Pinch-off condition in JFET? Explain with diagram. CLO1 3

(c) Explain the working principle of MOSFET. CLO1 3



Answer any four (04) out of the following questions:

$4 \times 10 = 40$

[The figures in the right margin indicate full marks. Symbols are used have their usual meaning.]

No.	Questions	CLOs	Marks
Q1.	(a) Define complex number.	CLO1	[1]
	(b) For any complex numbers $z_1, z_2, z_3, \dots, z_n$. Show that $ z_1 + z_2 + z_3 \leq z_1 + z_2 + z_3 $.	CLO2	[3]
	(c) State De Moiver's theorem and find all values of the following: (i) $(1+i)^n + (1-i)^n = 0$. (ii) $(1+i)^n + (1-i)^n = 0$.	CLO3	[6]
Q2.	(a) Define complex and analytic function.	CLO1	[1]
	(b) State and prove the necessary conditions for a function $f(z)$ to be analytic in a region.	CLO2	[5]
	(c) Show that the function $f(z) = \begin{cases} \frac{x^3y^4(x+iy)}{x^6+y^8}; & \text{when } z \neq 0 \\ 0; & \text{when } z = 0 \end{cases}$, are not analytic at the origin, although Cauchy-Riemann equations are satisfied.	CLO3	[4]
Q3.	(a) Define harmonic and harmonic conjugate function.	CLO1	[2]
	(b) Show that the real and imaginary parts of an <u>analytic</u> function are harmonic function.	CLO2	[4]
	(c) Show that the function $u = 3x^3 - 9xy^2 + 5x - 6y + 3$ is a harmonic function. Also find harmonic conjugate of u .	CLO3	[4]
Q4.	(a) Define complex integration.	CLO1	[2]
	(b) State and prove Cauchy's integral formula.	CLO2	[4]
	(c) Find $\oint_c \frac{\sin^6 z}{(z-\frac{\pi}{6})^3} dz$, where c is the circle $ z = 1$.	CLO3	[4]
Q5.	(a) Define branch point and contour.	CLO1	[2]
	(b) If we choose the principal branch of $\tanh^{-1} z$ to be that one for which $\tanh^{-1} 0 = 0$. Prove that $\tanh^{-1} z = \frac{1}{2} \ln \left(\frac{1+z}{1-z} \right)$.	CLO3	[4]
	(c) Calculate $\int_2^{4-2i} (z^2 - 4iz + 6) dz$.	CLO3	[4]
Q6.	(a) State Taylor and Laurent series.	CLO1	[2]
	(b) Expand $f(z) = \ln \left(\frac{1+z}{1-z} \right)$ using Taylor series expansion at $z = 0$.	CLO3	[4]
	(c) Determine the Laurent series for $f(z) = \frac{3}{z+6}$ that are valid in the region $ z-4 < 6$.	CLO3	[4]

Faculty of Science, Engineering and Technology
Department of CSE/EEE/Math
CSE 121: Structured Programming Language

Batch: 25th

Section: 1/2/3/4

Date: 30/09/2024

Final Examination

Semester: Spring 2024

Time: 2 Hours

Total Marks: 40

Answer five questions of the followings.

$5 \times 8 = 40$

- | | CLOs | Marks |
|--|------|-------|
| 1. a) Write equivalent logical expressions of the following without negation: | CLO4 | 2 |
| i. $!(a > b)$ | | |
| ii. $!(a \leq b \ \&\& \ c \leq d)$ | | |
| iii. $!(a+1 == b+1)$ | | |
| iv. $!(a < 1 \ \ b < 2 \ \&\& \ c < 3)$ | | |
| b) Consider you are given an integer 'n' and a digit 'd'. Write a program in C to build another integer 'r' where 'd' is inserted in front of n. For example if n=532 and d=8, then r will be set to 8532. | CLO4 | 6 |
| 2. a) Briefly explain storage classes in C. | CLO3 | 3 |
| b) Write down a function that print all distinct elements within a set of integers within an array 'A' with respect to another set of integers within an array 'B'. The size and the elements of both arrays will be input to your program. Assume that all elements are unique within an array. For example id A={15,7,9} and B={1,2,3,15}, your program shall print 7,9. | CLO4 | 5 |
| 3. a) Write the basic structure of Union. | CLO3 | 2 |
| b) Write C functions for the following tasks using pointers: | CLO4 | 6 |
| i. Copying one string to another | | |
| ii. Comparing two strings | | |
| 4. a) Write the differences between Structure and Union. | CLO3 | 2 |
| b) Create a structure named Book to store book details like title, author, and price. Write a C program to input details for three books, find the most expensive and the lowest priced books, and display their information. | CLO4 | 6 |
| 5. a) Illustrate a flow chart a given number is Armstrong or not. | CLO2 | 2 |
| b) What is recursion in C? Write a program in C to print n th term of the Fibonacci series using recursive function. | CLO4 | 6 |
| 6. a) What are the basic operations of file in C? | CLO3 | 2 |
| b) Explain the all mode of files in C. | CLO3 | 6 |

char s1[50], s2[50];
 int *p1 = &s1[0];
 *p2 = &s2[0];
 strcmp(*p1, *p2);

int n;
 int f1 = 0;
 int f2 = 1;
 int f3 = 1;

7. a) Define pointer with example.
b) consider the following program:

```
#include<stdio.h>
int swap(int a,int b)
{
    int t=a;
    a=b;
    b=t;
}
int main()
{
    int x,y;
    printf("Enter the value of a and b: ");
    scanf("%d %d",&x,&y);
    printf("Before swapping value of a and b is %d and %d\n",x,y);
    swap(x,y);
    printf("After swapping value of a and b is %d and %d\n",x,y);
    return 0;
}
```

Write the output of the given program. Rewrite the program using pointers with output.



HAMDARD UNIVERSITY BANGLADESH

Department of Computer Science & Engineering
Faculty of Science, Engineering and Technology
BSC in Computer Science & Engineering
ENG 127: English Skills
Final Examination
Spring 2024

25th Batch
25 September 2024

2nd Semester
Time: 2 Hours

Total Marks: 40

Questions

1. Answer any one of the following:

- Paragraph terminator and its characteristics.
- What is paragraph? Discuss different components of paragraph.

CLOs	Level	Marks
CLO3	Underst and	05

2. Rewrite the following passage into its correct order:

- The ants quickly gathered food and stored it in their colony. 2
- One summer day, a grasshopper saw a group of ants working hard. 1
- The grasshopper, who had been singing all day, laughed at them for working so much. 3
- The ants told the grasshopper that they were preparing for winter. 4
- When winter arrived, the grasshopper had nothing to eat and regretted not working. The ants, however, had plenty of food to last through the cold months. 5

CLO2	Analyze	05
------	---------	----

3. Paraphrase the following passage:

On cold, wet morning, my class was filled with excitement, because someone had discovered that the next day was our teacher's birthday. Our teacher was the kindest person that ever existed. Everyone wanted to get him a present. I, very much wanted to show my appreciation too. That day, I spent the whole afternoon shopping for a present. After a long search, I finally made up my mind to give him a book which he received with pleasure.

CLO2	Analyze	05
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4. Write down topic sentence from the given titles:

- The Uses and Abuses of Mobile Phone
- Natural disaster
- Dowry system
- How to improve the English Skills
- My university campus

CLO4	Apply	5x1=5
------	-------	-------

5. Write a review of a book that you have recently read (in approximately 250-300 words).

CLO4	Analyze	10
CLO5		

6. Write a paragraph on the given topic (within 150-200 words):

'General Education vs Technical Education' ✓

or

'Anti-Quota Movement in Bangladesh'

CLO4	Create	10
CLO5		

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 122: Structured Programming Language

Batch: 25th
Section: 1
Date: 18/09/2024

Semester: Spring 2024
Time: 2 Hours
Total Marks: 40

Lab Final Examination

Part A: Code Writings

Time: 1 Hour
(11.00 am – 12.00 pm)

Marks: 20

1. The factorial function is used frequently in probability problems. The factorial of a positive integer n (written $n!$ and pronounced “ n factorial”) is equal to the product of the positive integers from 1 to n . Write a program that evaluates the factorials of the integers from 1 to 5. 10

2. The simple interest on a loan is calculated by the formula 10
$$\text{interest} = \text{principal} * \text{rate} * \text{days} / 365;$$

The preceding formula assumes that rate is the annual interest rate, so it divides by 365 (days per year). Develop a program that uses `scanf` to input principal, rate and days for three (3) people's loans, and will calculate and display the simple interest for each loan, using the preceding formula.

3. Write a program in C to find a given number is Prime or not. 10

4. Write a program in C to find a given number is Palindrome or not. 10

Part B: Output Writings
(Implementation part)

Time: 45 minutes
(12.15 pm – 01.00 pm)

Marks: 20

1. Write a function to return an array containing all elements common to two given arrays containing distinct positive integers. You should not use any inbuilt methods. You are allowed to use any number of arrays. 15

The signature of the function is:
`int fun(int[] first, int[] second)`

Examples:

if the input parameters are	return
<code>{1,8,3,2},{4,2,6,1}</code>	<code>{1,2}</code>
<code>{1,3,7,9},{7,1,9,3}</code>	<code>{1,3,7,9}</code>
<code>{1,2},{3,4}</code>	Not matched

part (B) 1220
Union part (B)

2. Write a program in C to print the maximum and minimum value of an array.

05

3. Write a program in C to calculate the number of vowels of a given string.

05

Hamdard University Bangladesh

Dept. of EEE, CSE and Mathematics
Lab Final Examination (Spring'24)

Course code and Title:
Time: 1-hour

02
Marks: 40

Answer all the Questions from the following sets of questions

1. Write down the name of the experiments you have performed in your physics lab. 03
2. Suppose, you have a sphere and a cube. Now, for measuring the length of the cube, you found the main scale reading of 5 cm and Vernier scale reading of 0.3 cm. The diameter of the sphere is the last two digits of your Student ID (unit in mm). Suddenly, you insert the sphere into the cube. Find the volume of the void portion after inserting the sphere into the cube. 05
3. Analyze the time period vs distance figure of compound pendulum for the measurement of gravitational acceleration. 08
4. What type of problems you faced in your lab? 02
5. Write the formulae of the following terms and denote the parameters: 05
 - a) Radius of curvature of spherical surface
 - b) Rigidity modulus of a spring
 - c) Rigidity modulus of a wire
 - d) Refractive index of liquid
 - e) Vernier constant and pitch
6. Fill the appropriate answer in the blanks given below: 03
 - (a) The value of g in ideal case is cms^{-2}
 - (b) The no. of divisions of Vernier scale is 100. The VC is.....
 - (c) If we connect three identical springs in series with a constant of K the, the equivalent spring constant is.....
7. Mark the following statements as "T" for True and "F" for False. 03
 - (a) T^2 vs L graph of a simple pendulum is parabolic.
 - (b) m vs k graph of a simple pendulum is straight line.
 - (c) Negative error of a slide calipers is always subtractive.
8. Write down the correct answer from the Multiple-Choice Questions given below: 04
 - (a) Vernier calipers helps in measuring-
 - (i) External diameter
 - (ii) Internal diameter
 - (iii) Thickness
 - (iv) All of them
 - (b) Spring constant is dependent on-
 - (i) mass
 - (ii) the value of g
 - (iii) material of spring
 - (iv) All of them
 - (c) If a spring is vibrated in a downward lift, then, the value of g will be-
 - (i) increased
 - (ii) decreased
 - (iii) same as before
 - (iv) none of the above

M-1 VxV.C D
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179

(d) What do you think about me? -

- (i) Bad
- (iii) Worse

- (ii) Very bad
- (iv) Worst

9. Viva and lab performance